

Final Math Block: FWA vs Newtonian GR

Newtonian Limit of GR

Poisson equation:

$$\nabla^2 \Phi = 4\pi G \rho_b$$

For spherical symmetry:

$$d\Phi / dr = G M_b(r) / r^2$$

If baryon mass is finite as $r \rightarrow \infty$ ($M_b(r) \rightarrow M_0$):

$$d\Phi / dr \sim G M_0 / r^2$$

$$\Rightarrow \Phi(r) \sim -G M_0 / r$$

For circular velocity:

$$v^2(r) = G M_0 / r$$

$$\Rightarrow v(r) \propto r^{-1/2}$$

Observation

Flat rotation curves indicate:

$$v(r) \rightarrow v_0 \neq 0 \quad (r \rightarrow \infty)$$

From $v(r) = v_0$ it follows:

$$|d\Phi / dr| = v_0^2 / r$$

$$\Rightarrow \Phi(r) \sim v_0^2 \ln r + C$$

Flat curve = logarithmic potential.

Effective Density

Substituting $\Phi(r)$ into Poisson operator:

$$\nabla^2 \Phi = (1/r^2) d/dr (r^2 d\Phi/dr) = v_0^2 / r^2$$

To satisfy GR, an effective density is required:

$$\rho_{eff}(r) = v_0^2 / (4\pi G r^2)$$

The Dichotomy

1. **Add Source (Λ CDM):** Introduce Dark Matter profile $\rho_{DM} \propto r^{-2}$.
2. **Modify Operator (FWA):** Postulate $\nabla^2 \Phi \neq 4\pi G \rho_b$.

FWA Realization

FWA introduces the fundamental field equation:

$$E_{FWA}[\Phi, Lu, \eta_{coh}] = 0$$

Which asymptotically yields:

$$\Phi(r) \sim v_0^2 \ln r$$

directly from E_{FWA} without ρ_{DM} .

Conclusion

GR + Baryons and data are strictly inconsistent without choosing Option 1 or Option 2.